

B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

CE 19-201-0602 DESIGN OF STEEL STRUCTURES (2019 Scheme)

(Use of IS 800:2007, IS 801: 1975, IS 811: 1987 and steel tables are permitted)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Apply the provisions in Indian standard codes of practice for the design of various steel structures.
CO2: Design connections, tension and compression members.
CO3: Design beams and plate girders.
CO4: Design light gauge steel structures.
CO5: Design of continuous beams and simple frames using the concept of plastic design.
CO6: Design of structural components using timber.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

I.

Design the following joints using bearing type bolts between two plates of width 200 mm and thicknesses 10 mm and 18 mm respectively to transmit a factored load of 150 kN:

- Lap joint
- Single cover butt joint with cover plate of 8 mm.
- Double cover butt joint with 8 mm covers plates. Use plates made of Fe 410 grade steel and 16 mm diameter bolt of grade 4.6.

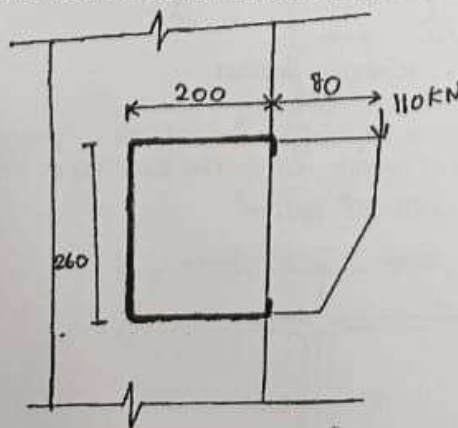
Marks	BL	CO	PO
18	L4	1,2	2,3

(12)

OR

- II. (a) Explain different types of welded connections.
(b) A bracket plate is welded to the flange of a column section ISHB300 @ 618N/m. Calculate the size of the weld required to support a factored load of 110 kN. Assume shop welding.

5	L1	2	1
13	L3	1,2	2



(P.T.O.)

III.

Design a bridge truss diagonal subjected to factored tensile load of 350 kN. The length of diagonal is 3 m. The tension member is connected to gusset plate 14 mm thick with one line of 20 mm diameter bolt of grade 8.8.

18 L4 1,2 2,3

OR

IV. (a) Design a single angle discontinuous strut to carry factored axial compressive load of 60 kN. The length of strut is 3 m. It is connected to 10 mm thick gusset plate by 20 mm diameter 4.6 grade bolt. Use Fe 410 steel.

10 L4 1,2 2,3

(b) Design a column to support factored load of 1040 kN. The column has an effective length of 7 m with respect to Z axis and 5 m with respect to Y axis.

8 L4 1,2 2,3

V. (a) Explain in detail the design procedure for the design of laterally supported beams.

6 L2 1,3 1

(b) Design a laterally unsupported beam for the following data.
Effective span = 4 m

12 L4 1,3 2,3

Maximum bending moment = 600 kNm.

Maximum shear force = 250 kN. Use Fe 410 grade steel.

OR

VI.

Design a welded plate girder of 20m using the tension field action for the following factored forces.

18 L4 1,3 2,3

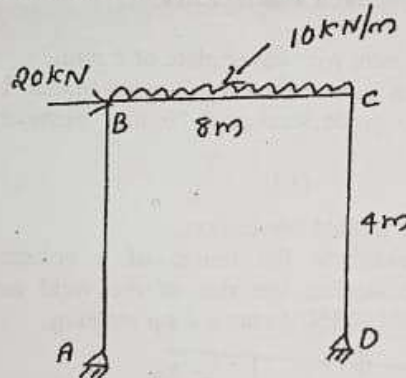
Maximum moment $M_z = 5500$ kNm.

Maximum shear force = 990 kN.

The girder is laterally restrained. Connections need not be designed.

VII. (a) Find the plastic moment for the given frame.

12 L3 5 2



(b) Find the shape factor for a triangular section.

6 L2 5 2

OR

VIII.

Compute the permissible axial load for the column section shown in figure. Effective length of column is 5 m. The steel has an yield point of 3550 kg/cm^2 and $E = 2.03 \times 10^6 \text{ kg/cm}^2$.

18 L3 4 2

